US Natural Disaster Temporal Analysis & Visualization

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1. Project description

The **US Natural Disaster Temporal Analysis** project is a data-driven study designed to analyze how disaster declarations evolve over time across the United States. The primary goal is to identify patterns, trends, and seasonality in the data to understand the frequency and nature of these events.

Key Objectives:

- **Temporal Trends:** Analyze how total disaster declarations change from year to year.
- **Incident Comparison:** Compare the evolution of different disaster types (e.g., Fire, Flood, Hurricane) over time.
- **Seasonality:** Identify recurring monthly patterns to determine which times of the year are most prone to specific disasters.
- **Data-Driven Insights:** Provide justifications for sudden spikes, drops, or increasing trends observed in the historical record

2. Dataset Used

The project utilizes the **Federal Emergencies and Disasters,1953-Present**, a comprehensive collection of disaster declaration records. This dataset is essential for benchmarking temporal shifts in environmental crises. even though the dataset says present it was last updated in the year 2017, So the end data we can find in the dataset is from the year 2017.

Dataset Characteristics:

- **Temporal Data:** Contains detailed date columns which are validated and converted into datetime format for analysis.
- **Categorical Data:** Includes incident types, allowing for the grouping and aggregation of specific disaster categories.
- **Granularity:** Data allows for extraction of specific time-based features, including the **Year** and **Month** of each declaration.

3. Environment Setup

The analysis is implemented in **Python** using **Visual Studio Code (VS Code)**. To ensure data integrity and efficient processing for the visualization tasks, the following tools were utilized:

- **Pandas:** Used for loading the dataset, validating time-related data, and performing aggregations by year and month.
- **Matplotlib:** Used as the primary visualization library to generate line and bar charts for disaster trends.
- **Jupyter Notebook:** Employed within VS Code to maintain a structured, cell-based workflow for code execution and data exploration.

Following are the requirements we have installed for the complete project but for this specific milestone we have used only Pandas, Matplotlib and Jupyter Notebook.

```
numpy
pandas
matplotlib
seaborn
plotly
geopandas
folium
jupyter
```

4. Data Exploration s Validation

Before generating any visualizations, a comprehensive exploration phase was conducted to ensure the dataset met the structural requirements defined in the project workflow.

4.1. Verifying Time Data Integrity

The validity of the entire temporal analysis depends on the accuracy of the date columns. The following checklist was completed:

- **Format Validation:** Confirmed that the date column is in the standard datetime format to allow for time-series operations.
- **Feature Extraction:** Successfully extracted granular time features, specifically **Year** and **Month**, to enable yearly and seasonal grouping.
- **Missing Value Analysis:** Checked for null or invalid date entries that could lead to abnormal drops in the trend lines.
- **Column Name Validation:** We have renamed the disaster type column to incident type that it matches data according to the milestone-2.

4.2. Initial Data Summary

A sample of the dataset was loaded to visually confirm the shape and quality of the data entries. This ensured that the incident types (e.g.. Fire, Flood, Hurricane) were correctly labeled and ready for aggregation.

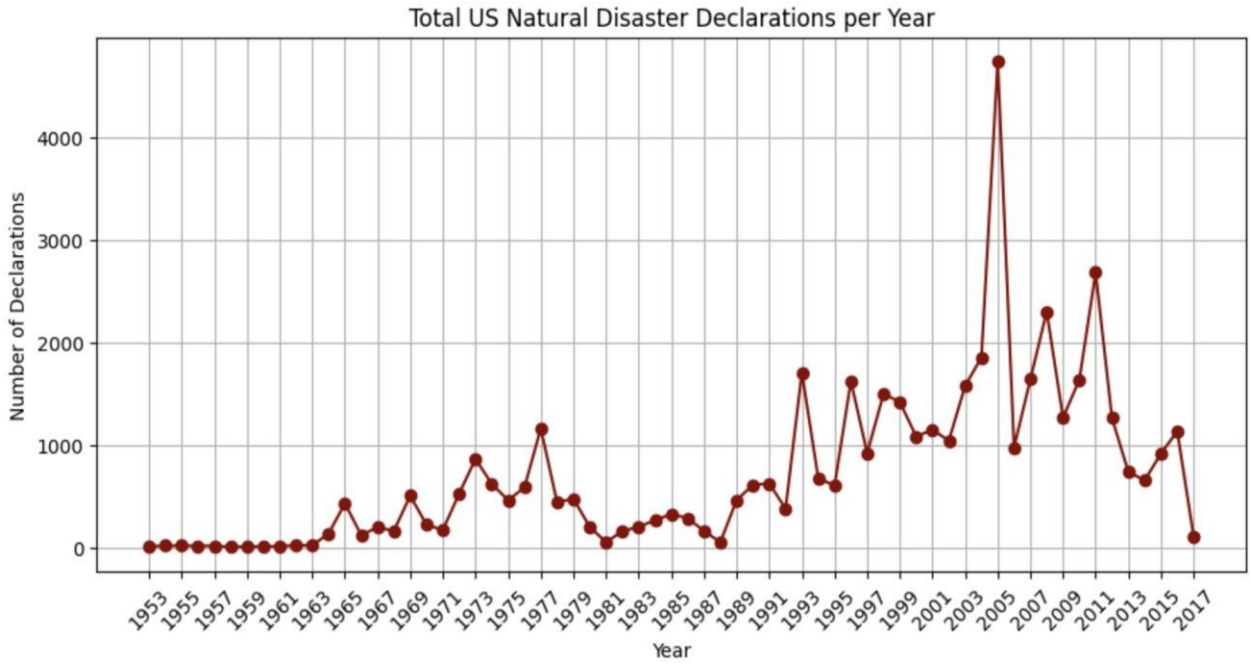
5. Temporal Trend Analysis

The primary focus of this phase is to analyze how disaster declarations evolve over time. By identifying patterns, trends, and seasonality, we can justify specific spikes or drops in the historical data.

5.1. Yearly Trend Analysis

The first step in understanding the temporal evolution was to group the data by year and count the number of disaster declarations.

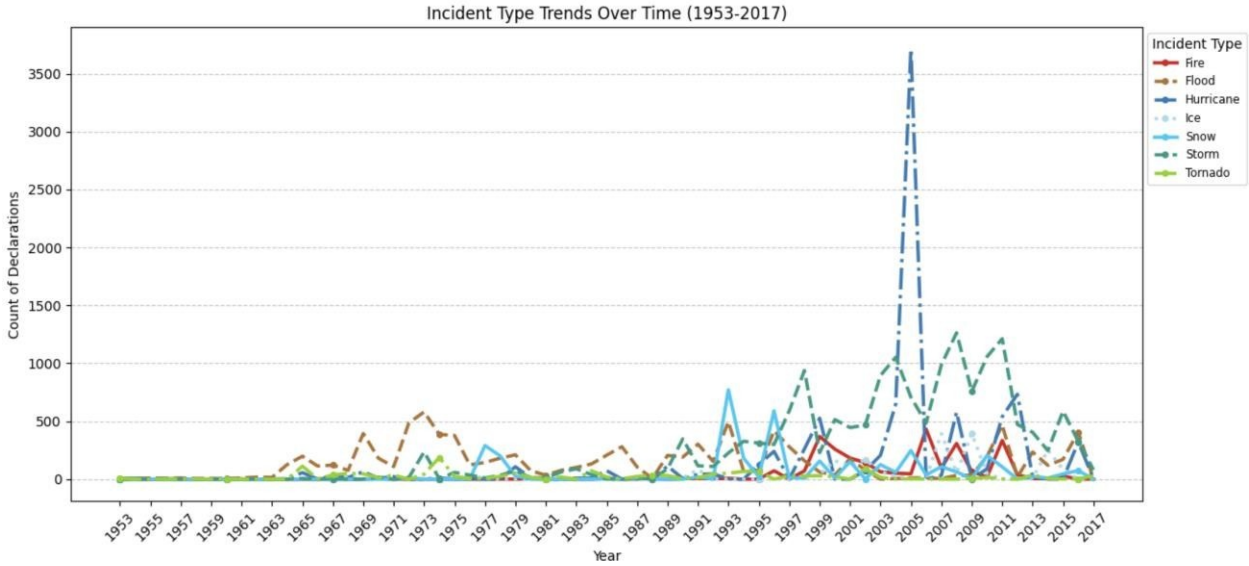
- **Task:** Aggregate total disasters per year and visualize them using a line chart.
- **X-Axis:** Year.
- **Y-Axis:** Number of Declarations.
- **Observation:** We can observe that the number of declarations increased by time, which shows a consistent level of increase in declarations reported and organized by the Federal Emergency Management Agency.



5.2. Incident Type Trends Over Time

To gain deeper insights, we compared how different disaster types evolve relative to one another.

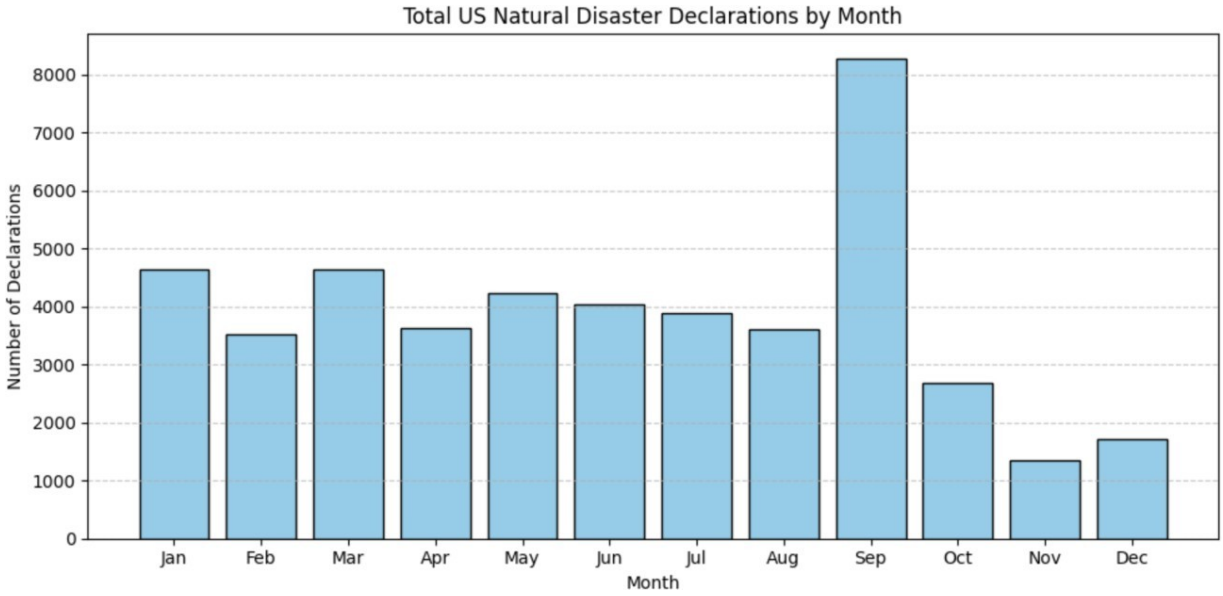
- **Task:** Group data by year and incident type to visualize multi-line comparisons.
- **Comparison:** This chart highlights which disaster types are the most frequent and which are increasing at the fastest rate.
- **Observation:** We can see the highest number of disasters declared over the timeline 1953-2017 and among the 20 disasters top 7 are hurricane, fire, flood, ice, snow, storm, tornado.



5.3. Seasonality Analysis (Monthly Trends)

Beyond long-term yearly growth, this task identifies recurring patterns within a standard calendar year.

- **Task:** Grouped the dataset by **month** and performed a count of disasters per month to identify seasonal peaks.
- **Visualization:** A line chart (or bar chart) was generated with **Month** on the X-axis and **Number of Declarations** on the Y-axis.
- **Interpretation:** This analysis helps determine if disasters are seasonal and which specific months historically carry the highest risk of declarations.
- **Observation:** The data shows a recurring peak in disasters during the late summer months, which correlates with historical hurricane and wildfire seasons and the highest can be seen in september.



6. Advanced Time Insights

To refine the raw data and reduce annual noise, the following advanced analytical techniques were applied to the US Natural Disaster dataset:

6.1. Rolling Average

A moving average (window = 3 or 5 years) was applied to the yearly trend data.

- **Purpose:** This technique helps smooth out short-term fluctuations and makes long-term trends more visible and easier to interpret.

6.2. Peak Detection

By analyzing the aggregated counts, specific years with the highest disaster declarations were identified.

- **Purpose:** This allows for the investigation of specific anomalies or major historical events that caused record-breaking disaster numbers.
- We have found the top 3 peaks among all the years and they are **2005** with 4736 declarations, **2011** with 2680 declarations and finally **2008** with 2300 declarations.
- The year with maximum growth is **1989** with over 805% growth.

7. Visualization Standards

To ensure technical accuracy and readability, every plot generated in this analysis adheres to a strict set of visualization standards. These standards ensure that data insights are communicated clearly and professionally.

Mandatory Elements Included:

- **Descriptive Titles:** Every chart includes a clear title defining the specific trend being analyzed.
- **Axis Labeling:** Both X-axis and Y-axis are explicitly labeled with units (e.g., Year, Month, Count).
- **Legend Integration:** For multi-line charts, a legend is used to differentiate between various incident types.
- **Clutter Reduction:** Graphs are scaled properly to avoid overlapping data points and ensure font readability.

8. Conclusion

The temporal analysis phase of the US Natural Disaster Visualization project was completed successfully. By following the structured workflow in Milestone-2, we successfully validated the historical dataset and extracted meaningful time-based features.

Key outcomes of this phase include:

Trend Identification: Successfully identified long-term growth patterns and sudden spikes in disaster declarations.

Comparative Insights: Developed a clear understanding of how different disaster types evolve and which categories are increasing most rapidly.

Seasonal Mapping: Established a baseline for recurring monthly patterns, providing a foundation for future predictive modeling.

The combination of validated data, standard-compliant visualizations, and detailed insight writing ensures that this project provides a reliable overview of disaster trends in the United States.